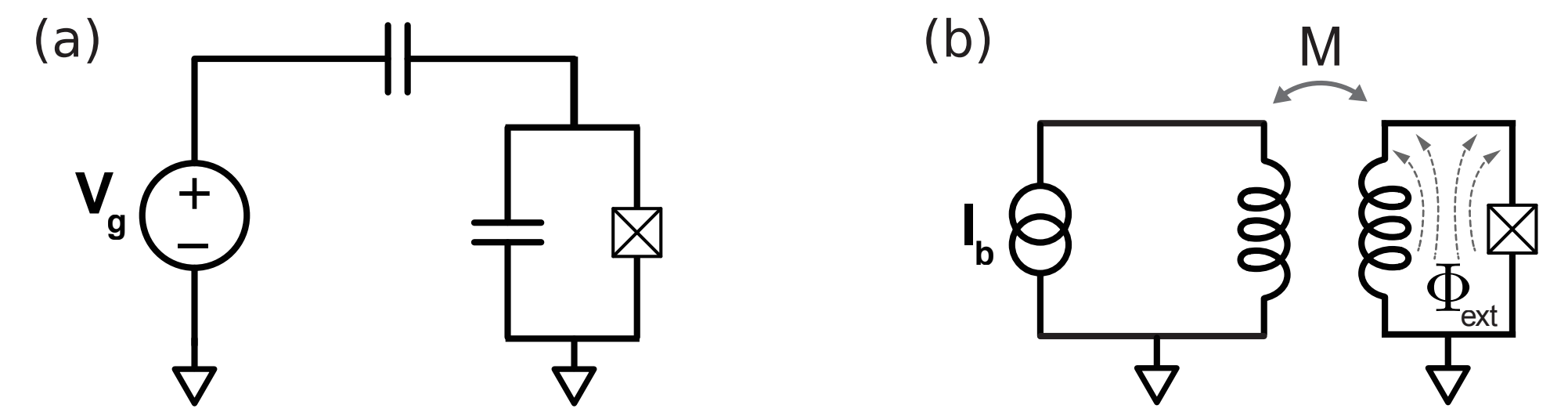

(a)
V_g
+
−
(b)
M
I_b
Φ_{ext}